

## Week 6 troubleshooting

| What they see                 | Cause                                     | Fix                                                                                              |
|-------------------------------|-------------------------------------------|--------------------------------------------------------------------------------------------------|
| health 404 after move         | route not registered / wrong method value | <code>r.GET("/health", h.Get)</code> — pass method, don't call <code>h.Get()</code>              |
| invalid memory address        | nil handler pointer                       | <code>h := handlers.NewHealthHandler()</code> before register                                    |
| package handlers not found    | wrong dir / module path                   | <code>internal/handlers</code> + import <code>student-management-system/internal/handlers</code> |
| <code>go test</code> no tests | wrong package path                        | from <code>backend/</code> : <code>go test ./internal/handlers</code>                            |
| import cycle                  | routes ↔ handlers wrong                   | handlers must not import the package that imports handlers for registration in a cycle           |
| FullName test fail            | trim/space rules                          | empty last → first only; trim both                                                               |
| still all code in main        | incomplete lab                            | move bodies into handler methods                                                                 |

### Minimal health handler

```
type HealthHandler struct{}

func NewHealthHandler() *HealthHandler { return &HealthHandler{} }

func (h *HealthHandler) Get(c *gin.Context) {
    c.JSON(http.StatusOK, gin.H{
        "status": "ok",
        "message": "Backend is running",
    })
}
```